Riverside Trunk Sewer Upgrade

Invitation to Tender — Synthetic Document for TunnelSpec AI Demo

1. Project Overview

This Invitation to Tender (ITT) covers the Riverside Trunk Sewer Upgrade, a 1,200 m segmental concrete tunnel of 1,800 mm internal diameter to be constructed by Earth Pressure Balance Tunnel Boring Machine (EPB-TBM) beneath the Riverside Quarter mixed-use development in central Birmingham. The Project is procured by Severn Trent Water under an NEC4 Option C target-cost contract. HB Tunnelling Ltd (HBT) is invited to bid as principal contractor, with City Council acting as highway authority and Network Rail as adjacent third-party asset owner along Chainage 600 m to 720 m.

The Works are split into six phases: Phase 1 Site Setup, Phase 2 Launch Shaft Construction, Phase 3 Tunnel Drive, Phase 4 Reception Shaft Construction, Phase 5 Pipeline and Lining Works, and Phase 6 Highway and Public Realm Reinstatement. The intended sequencing is strictly linear: Phase 3 depends on completion of Phase 2, and Phase 5 depends on completion of Phase 3.

2. Ground Conditions

The Geotechnical Interpretive Report (GIR) characterises the drive horizon as Mercia Mudstone Group overlain by 4 m to 6 m of River Terrace Gravels and made ground. Standard Penetration Test results across the route give N = 15 to 25 in the gravels and N > 50 in the weathered mudstone. The groundwater table is recorded at 4.0 m below ground level in piezometers BH-04 and BH-07, sitting within the gravels above the drive crown.

The combination of permeable gravels at the drive crown and a high groundwater table presents a credible risk of Groundwater Ingress at the TBM face during Phase 3 Tunnel Drive. This risk is the dominant geotechnical hazard for the Project. Tenderers must demonstrate adequate mitigation; the Employer's Requirements specify pre-drive ground treatment (cementitious permeation grouting) along Chainage 240 m to 360 m as the principal mitigation for groundwater ingress.

3. Risk Register — Geotechnical

Risk G-01: Groundwater Ingress at TBM Face. Likelihood: High. Severity: High. Triggered by encountering high-permeability gravels with hydraulic head above drive crown. Mitigated by pre-drive cementitious permeation grouting and by maintaining EPB face pressure at 1.4 to 1.8 bar.

Risk G-02: Surface Settlement. Likelihood: Medium. Severity: High. Settlement risk is closely related to groundwater ingress: dewatering at the face draws fines from the overlying gravels and mobilises consolidation settlement of the made ground. Mitigated by precise face pressure control and by the Settlement Monitoring Programme described in Section 6.

4. Risk Register — Programme & Operations

Risk P-01: TBM Stoppage. Likelihood: Medium. Severity: High. A sustained groundwater ingress event triggers TBM stoppage because the EPB cannot maintain face pressure once the spoil becomes excessively fluidised. Recovery requires intervention through hyperbaric chamber access, typically adding 10 to 20 working days per event.

Risk P-02: Programme Delay. Likelihood: Medium. Severity: High. TBM stoppage on the critical path triggers Programme Delay across Phase 3 Tunnel Drive and, by the linear dependency described in Section 1, into Phase 5 Pipeline and Lining Works. The Employer imposes liquidated damages of £25,000 per calendar day beyond Sectional Completion Date for Phase 3.

5. Risk Register — Third Party & Commercial

Risk T-01: Third-Party Building Damage. Likelihood: Low. Severity: High. Surface Settlement above the drive corridor can cause cosmetic and structural damage to the listed Edwardian terrace at 14-22 Riverside Walk. This risk is presently UNMITIGATED in the Employer's Requirements and tenderers are invited to propose a Party Wall Agreement strategy and compensation grouting allowance.

Risk C-01: Cost Overrun. Likelihood: Medium. Severity: High. Programme Delay triggers Cost Overrun through prolongation of site-establishment costs (Preliminaries running at £42,000 per week) and through liquidated damages exposure. Cost Overrun is also related to Third-Party Building Damage exposure should compensation grouting or remediation be required.

6. Mitigation Measures

M-01 Pre-Drive Ground Treatment. Cementitious permeation grouting along Chainage 240 m to 360 m. This mitigates Groundwater Ingress.

M-02 Settlement Monitoring Programme. A grid of automated total stations and precise levelling along the drive corridor, with trigger and action levels set at 5 mm and 10 mm respectively. This mitigates Surface Settlement.

Note that no mitigation is currently proposed for Third-Party Building Damage; tenderers are required to price and design this themselves.

7. Phase Dependencies & Required Materials

Phase 2 Launch Shaft Construction requires Sheet Piles (Larssen 605 grade), King-post propping, and a 6 m internal diameter excavation to 18 m depth. Phase 2 is the predecessor to Phase 3.

Phase 3 Tunnel Drive requires the EPB-TBM (Herrenknecht S-1010 class), Precast Concrete Segmental Lining (250 mm thick six-segment ring at 1.2 m pitch), Cementitious Tail Grout, and a Drive Length specification of 1,200 m. Phase 3 depends on Phase 2 and is the predecessor to Phase 5.

Phase 5 Pipeline and Lining Works requires the carrier pipe, internal secondary lining, and pressure-testing infrastructure. Phase 5 depends on completion of Phase 3.

8. Stakeholder Responsibilities

HB Tunnelling Ltd is responsible for Phase 2, Phase 3, Phase 4, and Phase 5 delivery. Severn Trent Water is responsible for Project sponsorship and acceptance testing. City Council is responsible for highway closure approvals during Phase 1 and Phase 6. Network Rail is responsible for possession management between Chainage 600 m and 720 m.

9. Programme Milestones

Sectional Completion Date for Phase 2: T0 + 16 weeks. Sectional Completion Date for Phase 3: T0 + 52 weeks. Sectional Completion Date for Phase 5: T0 + 70 weeks. Practical Completion: T0 + 80 weeks.

10. Bill of Quantities — Indicative Summary

Section A Preliminaries: site establishment £180,000, project management £210,000, temporary works design £75,000, insurance and bonds £95,000, environmental monitoring £60,000. Section B Shafts: launch shaft excavation and sheet piling £640,000, reception shaft excavation and sheet piling £420,000. Section C Tunnelling: EPB-TBM mobilisation and demobilisation £540,000, segmental lining supply £1,260,000, primary tunnel drive at £4,800 per linear metre over 1,200 m, tail grouting £180,000.

Section D Ground Treatment: cementitious permeation grouting along Chainage 240 m to 360 m £310,000, contingency for additional treatment if encountered £120,000. Section E Pipeline: carrier pipe supply and installation £490,000, internal secondary lining £180,000, hydrostatic pressure testing £60,000. Section F Reinstatement: highway reinstatement £215,000, public realm reinstatement £140,000. Section G Risk Allowances: contingency 8 percent of works value, design risk allowance 4 percent.

11. HSE & Environmental Constraints

Working hours are restricted to 07:00 to 19:00 Monday to Friday, with no weekend working permitted along Chainage 0 m to 240 m due to residential amenity. Noise emission at the site boundary is capped at 70 dB(A) during permitted hours. Dust suppression is required at the launch and reception shafts during excavation. The drive corridor passes within 30 m of the Riverside Walk Conservation Area and within 15 m of two protected London Plane trees, both of which are subject to Tree Preservation Orders.

Spoil management constraint: contaminated arisings from the made ground at the launch shaft must be characterised under WM3 guidance and disposed of to a permitted facility. The tendered rate must include contingent provision for hazardous classification. Discharge of any TBM process water to surface water is prohibited; tenderers must propose a closed-loop water recycling and treatment arrangement at the launch shaft.

12. Quality, Testing & Acceptance

Acceptance of the segmental tunnel lining is conditional on internal CCTV structural survey demonstrating no longitudinal cracking exceeding 0.3 mm and no offsets between rings exceeding 5 mm. Carrier pipe acceptance requires hydrostatic pressure testing at 1.5 times working pressure for two hours with permissible pressure loss not exceeding 2 percent. Tail grouting is to be verified by post-grout core sampling at 5 percent of rings, randomly selected.

Settlement monitoring data is to be reported daily to the Employer's Engineer during Phase 3 Tunnel Drive. Trigger Level breach (5 mm) requires written notification within 4 hours; Action Level breach (10 mm) requires immediate notification and triggers a joint review of TBM operating parameters. The Employer's Engineer reserves the right to instruct a temporary halt to the drive at Action Level until corrective measures are agreed.